MIL–R–5520C
30 JUNE 1964
SUPERSEDING
MIL–R–5520B
19 SEPTEMBER 1957

MILITARY SPECIFICATION

RESERVOIRS: AIRCRAFT, HYDRAULIC, NON-SEPARATED TYPE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification presents requirements for hydraulic fluid reservoirs (non-separated) wherein pressure is generated in the fluid by contact with gas. For separated reservoirs, see 6.3.

1.2 Classification. Hydraulic reservoirs shall be of the following types, as specified (see 6.2):

Type A—Nonpressurized
Type B—Pressurized

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN–P–515 Plywood, Container Grade
PPP–B–566 Boxes, Folding Paperboard
PPP–B–576 Box, Wood, Cleated, Veneer, Paper Overlaid
PPP–B–585 Boxes, Wood, Wirebound
PPP–B–591 Boxes, Fiberboard, Wood-Cleated
PPP–B–601 Boxes, Wood, Cleated-Plywood
PPP–B–621 Boxes, Wood, Nailed and Lock-Corner
PPP–B–636 Box, Fiberboard
PPP–B–640 Boxes, Fiberboard, Corrugated, Triple, Wall
PPP–B–676 Boxes, Set-Up, Paperboard

MILITARY

MIL–P–116 Preservation, Methods of
MIL–H–5440 Hydraulic Systems, Aircraft Types I and II, Design, Installation, and Data Requirements for
MIL–H–5606 Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL–P–7936 Parts and Equipment, Aeronautical, Preparation for Delivery
MIL–H–8775 Hydraulic System Components, Aircraft and Missiles, General Specification for
MIL–V–8813 Valves, Aircraft, Hydraulic Pressure Relief, Type II Systems
MIL–L–10547 Liners; Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL–STD–129 Marking for Shipment and Storage
MIL–STD–143 Specifications and Standards, Order of Precedence for the Selection of
MIL–STD–831 Test Reports; Preparation of

AIR FORCE-NAVY AERONAUTICAL

AN871 Flange—Straight Thread Boss Welding
AND10050 Bosses, Standard Dimensions for Gasket Seal Straight Thread

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 438 Age Controls for Synthetic Rubber Parts

(Copies of specifications, standards, publications, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1650

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill., 60606.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing (see 4.3).

3.2 Precedence. Hydraulic reservoirs shall conform to the requirements of this specification, the requirements of MIL–H–8775, and the requirements of MIL–H–5440, in that order of precedence. The choice and application of all other specifications and standards shall be in accordance with MIL–STD–143. In the event of conflict between the requirements of this specification and the requirements of any other specification, the above order of precedence shall be observed.

3.3 Data. Unless otherwise specified in the contract or order, no data other than reports and drawings accompanying preproduction samples (see 4.3.2) are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.4 Design and construction.

3.4.1 *Reservoir capacity.* Hydraulic reservoirs shall be designed to provide for the volume of fluid, gas space, and emergency reservoir capacity as specified herein. A diagram illustrating the capacity requirements is shown on figure 1.

3.4.1.1 *Fluid.* The design minimum volume of fluid shall be equal to the sum of the following:

(a) A sufficient amount of fluid to insure that the hydraulic pump suction is maintained for all attitudes and accelerations within the design limitations for which the reservoir is required to function. Assisted takeoff conditions, such as catapult launching of Naval aircraft, shall be taken into consideration in this phase of capacity.

(b) A fluid volume equivalent to 100 percent of the possible net depletion caused by cylinder volumetric changes which can occur in any ground checkout or flight condition, with reference to the normal reservoir filling attitude and position of the various services.

(c) A fluid volume equivalent to 100 percent of the reservoir fluid volumetric change caused by charging all accumulators to system operating pressure from a completely discharged position with no gas charge in the largest accumulator and a minimum gas charge in all others.

(d) A fluid volume equivalent to 130 percent of the volumetric capacity of the largest quantity-measuring type of hydraulic fuse in the system.

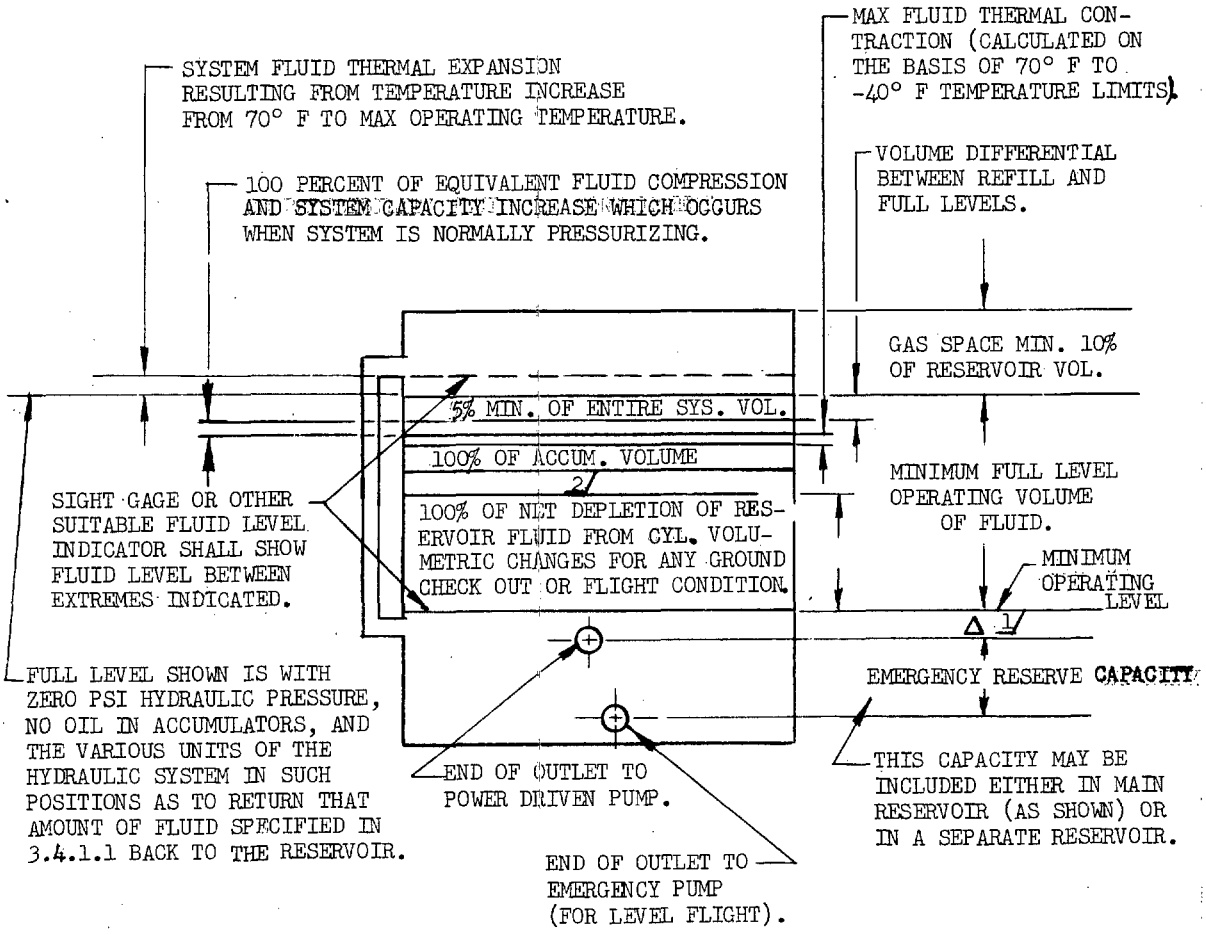
(e) That volume of fluid equivalent to the maximum thermal contraction which is expected to occur in the system. From 70° F filling temperature to −40° F is recommended as a basis for arriving at this volume.

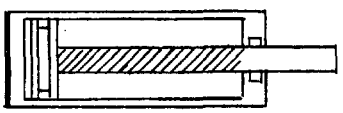
(f) A fluid volume equivalent to not less than 5 percent of the entire system fluid volume, including the reservoir, to minimize the normal frequency of filling. The differential volume between full and refill shall correspond to this leakage allowance.

(g) A fluid volume equivalent to that resulting from the effects of fluid compression, line and actuator expansion, and external seal deflection when the hydraulic system which the reservoir is serving is pressurized at normal operating pressure.

3.4.1.2 *Gas space.* Gas space shall be provided, which, together with other design features, will ensure that there will be no loss of fluid during normal operating conditions. The gas space volume shall be 10 percent (minimum) of the fluid volume as defined in 3.4.1.1. If the vent line (or the reservoir relief valve in the case of pressurized reservoirs) is not capable of accommodating a sufficient high rate of fluid flow during the most adverse surge condition obtainable without exceeding the reservoir design pressure, then the gas space must

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EXAMPLE OF VOLUMETRIC CHANGE IN AN ACTUATING CYLINDER AFTER ACTING THROUGH THE HALF CYCLE OF OPERATION WHICH RETURNS THE MOST FLUID TO THE RESERVOIR. EXCHANGE VOL. EQUALS VOL. OF THAT PART OF PISTON ROD INDICATED BY SHADING.

1/ THIS DISTANCE SHALL BE SUFFICIENT FOR BOTH THE GROUND SERVICING AND CRITICAL FLIGHT CONDITIONS (WITH THEIR CORRESPONDING MINIMUM OPERATING LEVELS) TO INSURE THAT HYDRAULIC PUMP SUCTION IS MAINTAINED.

2/ A VOLUME EQUIVALENT TO 130 PERCENT OF THE VOLUMETRIC CAPACITY OF THE LARGEST QUANTITY MEASURING TYPE HYDRAULIC FUSE IN THE SYSTEM.

Δ THESE DISTANCES MAY VARY BETWEEN GROUND AND FLIGHT CONDITIONS.

FIGURE 1. *Reservoir capacity requirements*

be made large enough to limit the maximum pressure to a value less than the design pressure. The surge conditions to be considered include those resulting from the acceleration of hydraulic actuators by aerodynamic loads or gravity due to actuation by emergency pneumatic systems. The condition of an overheated hydraulic system with its accompanying volumetric increase shall also be considered here.

3.4.1.3 *Emergency reserve capacity.* In cases where an emergency reserve capacity of fluid

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is required, the minimum volume shall be equal to 125 percent of the fluid required for emergency purposes or normal landing. The volume of units that can be operated by other than hydraulic means, provided that no fluid is drawn from the reservoir during such operation, need not be considered. No allowance for return of fluid to the reservoir shall be made in determining emergency reserve capacity. This emergency fluid volume shall be below the power pump outlet (for level flight) in cases where it is incorporated in the main reservoir; it may, however, be provided in a separate reservoir. In cases where the emergency fluid supply is incorporated in the main reservoir (as may be the case in trainer aircraft), the emergency supply shall be unaffected by the complete loss of the normal supply.

3.4.2 *Refill level.* The refill level shall be that level which results from lowering the fluid level from the full level by an amount equal to 5 percent of the entire system fluid volume, including the reservoir. If more than one full level is indicated on the instruction plate, each full level may have its corresponding refill level indicated to prevent the necessity for too frequent refilling. Each of these refill levels shall be determined as above.

3.4.3 *Interconnected auxiliary reservoirs.* When an auxiliary interconnected reservoir is filled through a primary reservoir, sufficient gas space must be provided in the primary reservoir for expansion in the auxiliary reservoir. The interconnecting supply line shall be of sufficient size to ensure that the auxiliary reservoir will completely fill in less time than it takes to fill the main supply reservoir under any operating temperature condition. If this interconnecting line is designed to function also in the capacity of a suction line during conditions in which flow leaving the auxiliary reservoir is returned to the primary reservoir, it shall conform with the suction-line flow requirements of MIL–H–5440. A check valve is to be provided in the line if necessary to prevent depletion of the oil supply in the auxiliary reservoir in inverted flight. The auxiliary reservoir vent line shall be so routed that there are no fluid traps in the line and it should be connected into the top of the main reservoir separately from the main reservoir vent line, to minimize loss of fluid overboard. In cases where an auxiliary or emergency reservoir is interconnected to a primary reservoir, it shall not be possible to lose the auxiliary or emergency fluid supply by a failure in the primary system, or vice versa.

3.4.4 *Return line inlet.* The inlet from the return line shall be so positioned that vortexing and foaming of fluid flowing into the reservoir will be kept to a minimum. The return line inlet shall be located below the minimum recommended fluid level, unless a suitable type antifoaming arrangement is used.

3.4.5 *De-aeration.* Means shall be provided to minimize surging, foaming, or vortexing of fluid as it enters the reservoir from the return line. The design shall prevent high velocity direct internal flow from the return inlet of filter element to the outlet to the power-driven pump.

3.4.6 *Filler unit.* A filler neck, at least 1 inch in diameter clear opening, shall be rigidly attached to the reservoir. Filling or replenishing the reservoir from a remote point by means of a hand pump or electric-driven pump, wherein all fluid entering the reservoir is filtered and the pump is a permanent part of the airplane installation, is acceptable in lieu of the "pour-in" method.

3.4.6.1 *Filler unit filter.* The filler neck shall be provided with a filter capable of removing from the replenishing fluid all particles larger than 0.0035 inch in size. This filter shall be capable of passing replenishing fluid into the reservoir at the rate specified herein. The design of the filler unit filter shall be such as to prevent damage in filling (caused by inserting a filling spout too deeply or improperly into the filler pipe), removal, or cleaning, and shall be secured in position in order not to be readily removable, except for cleaning purposes. A suggested filler unit mesh filter installation is shown on figure 2.

3.4.6.2 *Scupper.* A suitably designed scupper shall be provided at the filler port outside opening to drain spillage to the outside of the aircraft (see figure 2).

3.4.6.3 *Filling vent line.* A filling vent line of a type similar to that shown on figure 2 shall be provided to facilitate rapid filling and also to prevent overfilling by virtue of the location

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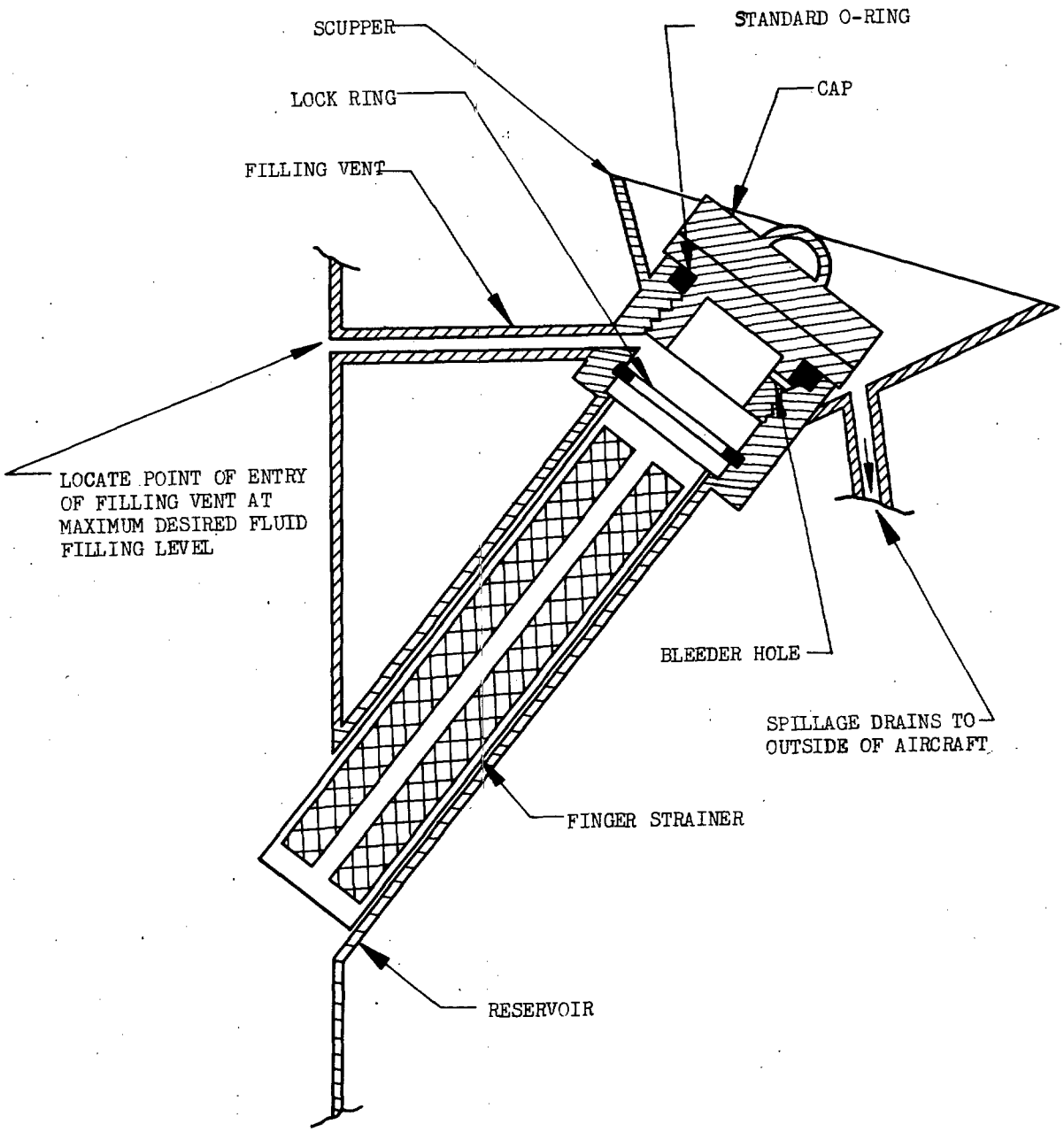


FIGURE 2. *Typical filler unit finger strainer*

of the point of entry of the vent line into the reservoir at the maximum desired fluid level.

3.4.6.4 *Filler cap.* A filler cap capable of preventing leakage of fluid or loss of pressure shall be provided. The filler cap shall be easily installed or removed by hand. Use of a filler cap assembly employing screw threads is not desired. Design of the cap shall provide for a minimum number of turns for complete cap disengagement. Provisions for automatic release of stored air pressure prior to complete filler cap disengagement shall be made. The filler cap assembly shall be designed to provide a minimum of 500 complete cycles of operation. A complete cycle shall consist of removal and installation of the cap. To prevent loss of the

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cap when removed, it shall be attached to the filler neck or reservoir with a chain or by other suitable means. The design shall be such that it will be readily apparent, upon visual inspection, that the cap is securely and properly installed.

3.4.7 *Fluid level indication.* Suitable means to give visual indication of fluid level in the reservoir between the lowest and highest operating levels (see figure 1) shall be provided. A dip stick shall not be used for fluid level indication.

3.4.8 *Sump.* The reservoir shall be provided with a sump volume below the lowest normal or emergency operating outlet with the aircraft in level flight or at rest.

3.4.9 *Drain.* Means shall be provided to ensure that reservoir and sump can be completely drained of fluid when the aircraft is at rest without undoing tubing connections. There shall be no cavities where fluid can be entrapped. Drain size shall permit complete draining in 5 minutes at −40° F.

3.4.10 *Structural strength.* Reservoirs shall have ample strength and rigidity to withstand 250 percent of maximum wrench torques required for making any and all tube connections without permanent deformation of the reservoir.

3.4.11 *Connections.* All fluid connections shall be made through flanged bosses conforming to AND10050, welded or otherwise suitably joined to the reservoir. Welding flanges bosses conforming to AN871 shall be used wherever applicable. Tubing welded to the reservoirs shall not be used except for filler necks and filling vents. The number of openings in the reservoir shall be kept to a minimum.

3.4.12 *Connection to atmosphere.* Means shall be provided which will transmit excessive internal reservoir pressure to atmosphere. When an open line is provided to relieve outside-to-inside differential pressure, a 10-micron nominal vent filler shall be provided.

3.4.12.1 *Design of connection to atmosphere.* Under conditions of maximum surge return flow to the reservoir, including emergency operation of hydraulic units, discharge of high-pressure air, or an overheated hydraulic system (such as may be caused from a continuously bypassing relief valve during system malfunctioning), and if sufficient air space is not provided whereby the maximum surge pressure is limited to an amount which is less than 67 percent of the reservoir proof pressure, the connection to the atmosphere (including relief valve, in the case of pressurized reservoir) shall be large enough to accommodate the flow of surging fluid without exceeding the aforementioned pressure. However, fluid which is lost overboard from this source shall not exceed that amount which is consistent with the requirements for safe flight of the airplane or the desired frequency of fluid replenishment. Provision may be made to bypass the vent filter element when the direction of flow is away from the reservoir. Means shall be provided to prevent excessive differential between internal and external pressures with quick increases in outside air pressure, as may be encountered in fast dives.

3.4.13 *Pressurization.* Provision shall be made on pressurized reservoirs to allow atmospheric air to enter the reservoir in the event of failure of the pressurizing source. This entry of atmospheric air shall occur before the inside pressure exceeds 0.5 psi less than the outside pressure.

3.4.14 *Filters.* Filtration shall not be provided as an integral part of the reservoir assembly without the prior approval of the procuring activity. If an integral filter is used, the filter element shall be replaceable without disconnecting tubing.

3.4.15 *Mounting.* Design of the reservoir shall provide for mounting in the aircraft in such manner that the reservoir will be securely fastened in place with no weight carried by the fittings or fluid lines attached thereto. The mounting shall cause no abrasion of the reservoir or its fittings and shall not impose undue strain on the reservoir. The mounting shall provide for easy removal of the reservoir from the aircraft.

3.4.16 *Marking.* Part number and port marking of the reservoir shall be in accordance with MIL–H–8775, except that the nameplate shall consist of the instruction plate specified herein.

3.4.17 *Reservoir instruction plate.* An identification and instruction plate (see figure 3) shall be attached to either the reservoir itself or to the immediately adjacent vehicle structure

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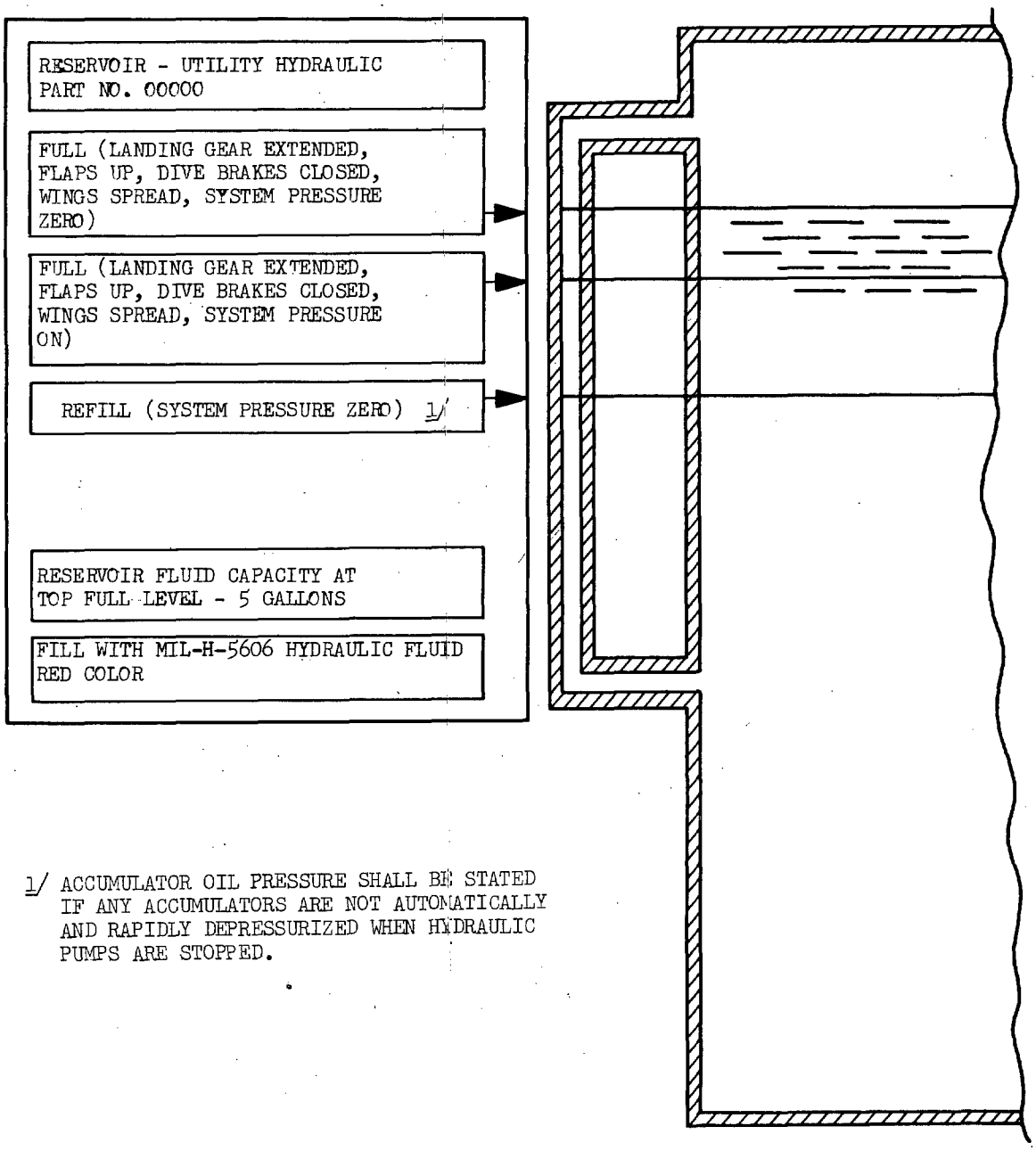
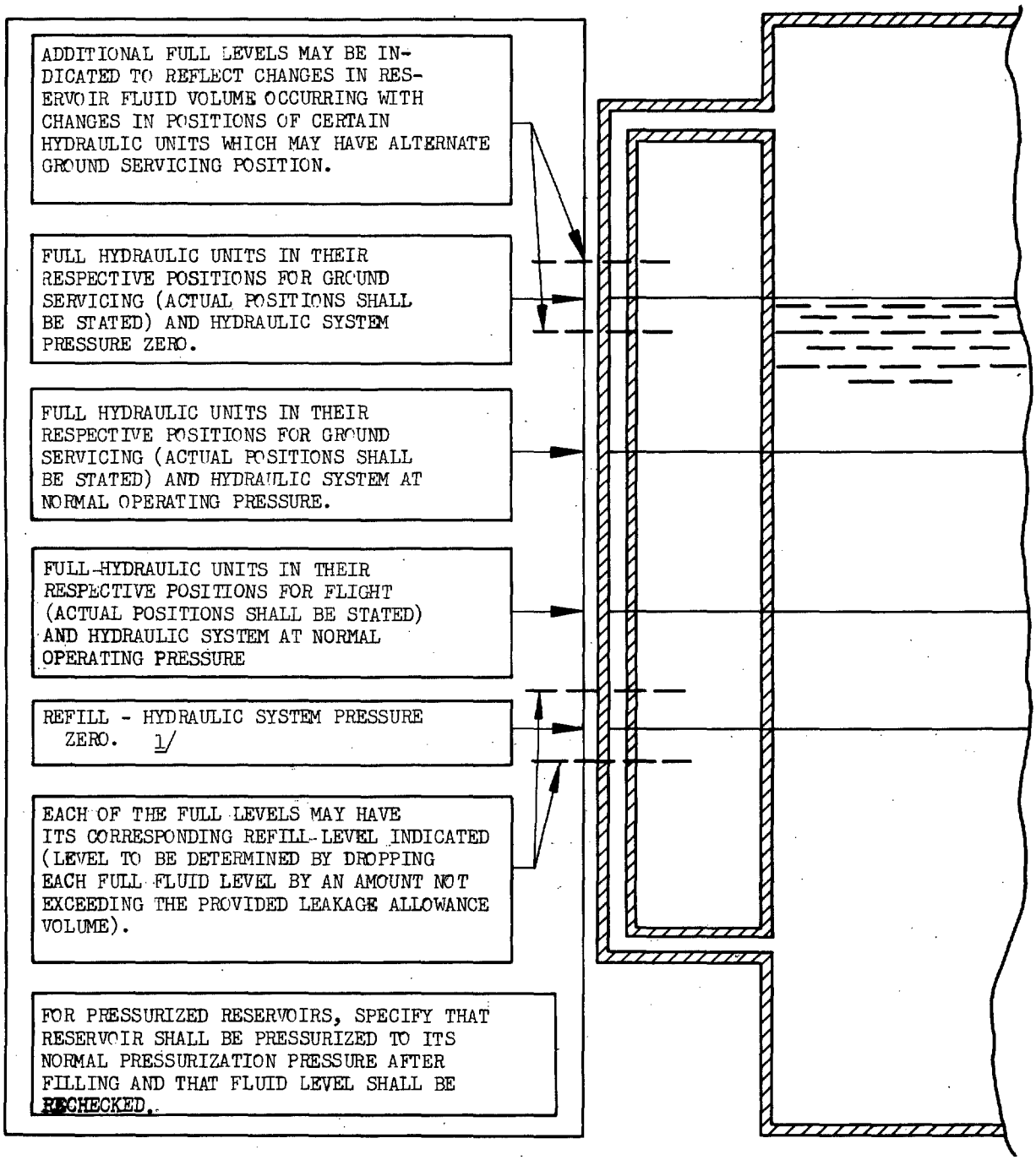


FIGURE 3. *Typical reservoir nameplate*

near the fluid level indicating device, positioned so as to be easily readable. This plate shall contain simple and brief, but complete, instructions for servicing the reservoir, and shall include at least the following minimum information:

(a) Name and part number of the reservoir.
(b) Specification and color of fluid, as applicable.
(c) Reservoir fluid capacity at full level.
(d) Full level indication reference.
(e) Refill level indication reference.
(f) Position of hydraulic devices and aircraft during filling procedure.

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1/ IF ONLY ONE REFILL LEVEL IS INDICATED, IT SHALL BE THAT LEVEL WHICH RESULTS FROM LOWERING THE FLUID LEVEL FROM THE UPPERMOST FULL LEVEL BY AN AMOUNT NOT EXCEEDING THE PROVIDED LEAKAGE ALLOWANCE VOLUME.

FIGURE 4. *Considerations for use in arriving at fluid levels to be designated on reservoir nameplates*

(g) Depressurizing procedure reference.

(h) Pressurization warning notice.

Additional full and refill levels under various conditions of system pressure, temperature, vehicle attitude, accumulator precharge, etc., shall be added as required (see figure 4). Any

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other special instructions may also be included.

3.4.18 *Age control.* Age control for any synthetic rubber parts contained in the design or installation shall be exercised in accordance with the intent and requirements of ANA Bulletin No. 438.

3.4.19 *Reliability.* Design and construction of the hydraulic reservoir shall be such as to ensure the maximum reliability. Due consideration shall be given to those factors which affect the reservoir reliability, including simplicity, number of parts, degree of handling required by personnel, etc. The design of all operating parts shall be analyzed for fail-safe characteristics.

3.4.20 *Strength at elevated temperature.* Pressurized reservoirs shall be capable of withstanding proof pressure at the maximum design temperature after exposure to elevated temperatures for the period of time specified in the manufacturer's drawing.

3.4.21 *Corrosion resistance.* All metals shall possess adequate corrosion-resistant characteristics or shall be suitably protected by use of permanent coatings to resist corrosion which may result from such conditions as dissimilar metal combinations, moisture, salt spray, and high-temperature deterioration. Hydraulic fluid contacting interior surfaces shall not be considered a suitable protection.

3.5 Performance.

3.5.1 *Filling time.* Filler unit design shall be such that the reservoir may be filled from empty to full in 5 minutes at 70° F.

3.5.2 *Operating characteristics.* With the reservoir fluid level at either full or refill, the fluid emerging from the reservoir outlets shall be free of detrimental quantities of froth or foam, and hydraulic pump supply shall be maintained for ground servicing and critical flight attitudes, respectively. Flight conditions should include the most severe sloshing expected from either rolling or pitching action as well as acceleration effects of assisted takeoff conditions. Foaming that occurs when the reservoir is tested in attitudes more severe than those for which it is required to operate shall disappear after the reservoir is righted within a length of time compatible with the design performance requirements of the particular hydraulic system which the reservoir is serving.

3.5.3 *Proof pressure.* Reservoirs shall withstand proof pressure as specified in 4.6.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examining and testing of the reservoirs shall be classified as:

(a) Preproduction inspections	(4.3)
(b) Quality conformance inspections	(4.4)

4.3 Preproduction inspections. Every hydraulic reservoir design submitted for acceptance in accordance with this specification shall be subjected to:

(a) The quality conformance inspections	(4.4)
(b) Negative pressure	(4.6.3)
(c) Capacity	(4.6.4)
(d) Immersion	(4.6.5)
(e) Relief valve operation	(4.6.6)
(f) Operating characteristics	(4.6.8)
(g) Vibration	(4.6.9)
(h) Endurance test	(4.6.10)
(i) Filling time	(4.6.7)

4.3.1 *Rejection.* Failure of any reservoir assembly to successfully comply with any of the requirements of the preproduction tests (4.3) shall be cause for rejection of that reservoir design.

4.3.2 *Preproduction sample.* One hydraulic reservoir assembly shall be used for the performance of the preproduction inspections. The use of machining and weldments in place of forgings, castings, etc., in the reservoir assembly used for testing shall require the prior approval of the procuring activity and shall be allowed only when substantiating data has been provided to demonstrate that differences in strength, fatigue properties, and other per-

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formance factors have been allowed for in the substitution. Test reports shall be in accordance with MIL–STD–831.

4.4 Quality conformance inspections. Every reservoir assembly submitted for acceptance under contract shall be subjected to:

(a) Examination of product ..... (4.6.1)
(b) Proof pressure ..... (4.6.2)

4.5 Test conditions.

4.5.1 *Fluid.* Unless otherwise specified, all hydraulic fluid used for testing shall be in accordance with MIL–H–5606.

4.5.2 *Temperature.* Unless otherwise specified herein, all temperatures referred to under 4.6 shall have a tolerance of ±5° F. Room temperature referred to herein shall be 70° to 110° F.

4.5.3 *Mounting.* The vibration test shall be conducted with the reservoir assembly mounted in a framework of structure corresponding to the proposed installation in the vehicle. Due consideration shall be given to shape, dimensions, material, strength, deflection characteristics, and any other circumstances applicable to the performance of the reservoir. This structural framework shall constitute the test cradle and shall be constructed in such a way as to be able to be rotated in any of its major axes as well as to be mounted on a vibration machine. All mounting provisions, tubing connections, etc., shall be identical to the intended vehicle installation as far as practicable. The cradle shall be capable of withstanding all environmental effects to which the test article will be exposed.

4.5.4 *Pressure.* Unless otherwise specified herein, all pressures referred to under 4.6 shall be those specified on the manufacturer's drawings.

4.6 Test methods.

4.6.1 *Examination of product.* The reservoir shall be inspected to determine compliance to this specification with respect to interchangeability, dimensions, materials, and workmanship.

4.6.1.1 *Packaging, packing, and marking.* Preparation for delivery shall be inspected for conformance to section 5.

4.6.2 *Proof pressure.* With the filler unit installed in the normal manner and all other ports and openings plugged, the reservoir assembly shall be filled and bled, and externally generated hydraulic fluid pressure shall be applied twice at one of the fluid ports for a period of 2 minutes each time. Minimum pressure shall be 200 percent of maximum design operating pressure in a type B design or 8 psig in a type A design. During and following the performance of the test there shall be no external leakage sufficient to form a drop. Slight wetting of the surface adjacent to any port shall not be cause for rejection. During the performance of the proof pressure test the reservoir assembly shall be measured before and after application of proof pressure with instruments calibrated to 0.001 of an inch. There shall be no indication of excessive permanent deformation. If any permanent deformation has occurred, the proof pressure test shall be repeated on the same test article and no further permanent deformation shall occur. Evidence of further permanent deformation after the second performance of the proof pressure test shall be cause for rejection of the reservoir design.

4.6.3 *Negative pressure.* The hydraulic fluid section of the reservoir shall be subjected to a pressure of 4 psi less than atmospheric ambient pressure. The negative pressure shall be maintained for a period of 5 minutes with no evidence of leakage, failure, or detrimental permanent deformation.

4.6.4 *Capacity.* A test shall be conducted to determine the following quantities to within ±1 percent:

(a) Total reservoir capacity, including both fluid and air volumes.
(b) Fluid capacity between the full level(s) and the outlet to the power-driven pump with the reservoir mounted in the attitude corresponding to normal aircraft ground attitude.
(c) Fluid capacity between the refill level(s) and the outlet to the power-driven pump with the reservoir mounted in the attitude corresponding to normal aircraft ground attitude.
(d) Fluid capacity between the levels of the power-driven pump outlet and the emergency pump outlet with the reservoir in a level flight attitude (applies only to reservoirs which include both

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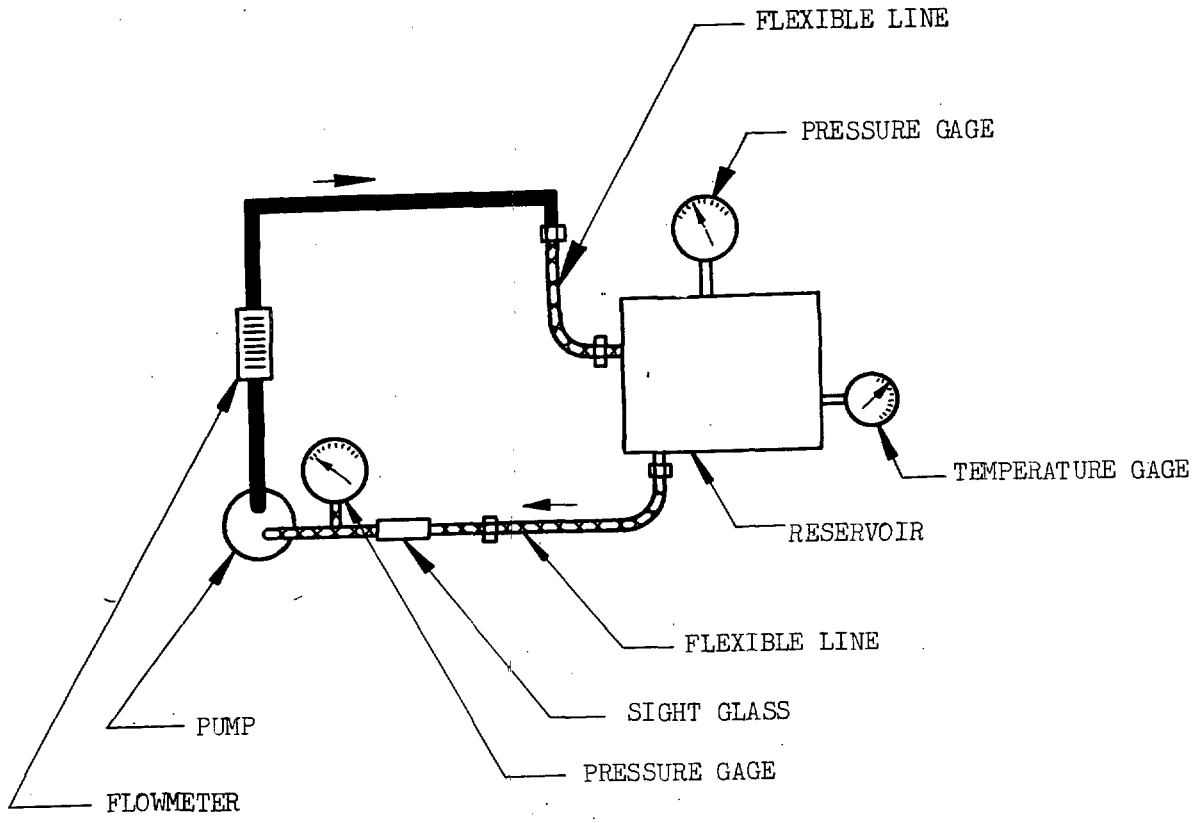


FIGURE 5. *Hydraulic reservoir operating characteristics test setup*

normal and emergency hydraulic fluid capacity in one reservoir).

4.6.5 *Immersion.* Reservoirs containing nonmetallic parts shall be subjected to an immersion test in accordance with the methods specified in MIL–H–8775.

4.6.6 *Relief valve operation.* If a pressure relieving device is an integral part of the reservoir assembly, the valving element shall be tested in accordance with applicable requirements of MIL–H–8775 and MIL–V–8813, except that the pressure used for cracking, full flow, and reseat shall be that specified on the manufacturer's drawings.

4.6.7 *Filling time.* The reservoir shall have all ports sealed, except the filler neck and vent lines, and shall be dry. A vent configuration conforming to that of the aircraft shall be attached to the reservoir. The reservoir shall be held in the attitude which it would assume when installed in an aircraft standing in the normal ground attitude. Hydraulic fluid at a temperature of 70° F shall be poured into the filler pipe at a rate sufficient to maintain the liquid level at the top of the filler pipe. Filling time under these conditions shall not exceed 5 minutes.

4.6.8 *Operating characteristics.* The reservoir shall be installed in a test setup similar to that shown on figure 5. The pump supply line configuration shall simulate that of the aircraft as closely as test conditions will permit. If the unit is of the pressurized type, the design pressurization shall be applied by means simulating the method used in the aircraft. Hydraulic pump supply characteristics at maximum design hydraulic flow shall be investigated for ground servicing and critical flight attitudes with fluid level adjusted to produce depletion in amounts as specified in 3.4.1.1. If any interruption of normal pump operation or output occurs, the attitude at which this interruption occurs and the time and conditions required for normal operation to be restored shall be determined and recorded. Fluid temperatures shall be recorded, and appearance of froth or foam and time required to disappear shall be observed and recorded.

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4.6.9 *Vibration.* The test reservoir, mounted in the test cradle described in 4.5.3, shall be fastened to a vibration machine capable of producing a rotary displacement and frequency of vibration specified herein, while being rocked in a manner to produce the sloshing specified herein. If the reservoir is type B, design pressurization shall be applied and maintained throughout the vibration test. The reservoir shall be filled to two-thirds of its fluid capacity with hydraulic fluid of the type to be used in this system (MIL–H–5606). The axis of vibratory motion of the test sample shall represent the shaft axis of the aircraft engine in the intended application. For testing reservoirs to be used in aircraft propelled entirely or partially by a reciprocating engine(s), the rotative frequency of the test assembly shall be 90 percent +0, −3 percent, of the normal rated crankshaft speed of the engine used. For testing all other aircraft reservoirs, the rotative frequency of the test assembly shall be 2,000 rpm +0, −60 rpm. The vibration machine shall be adjusted to produce a total displacement of $\frac{1}{32}$ inch +0.010, −0.000 inch, measured at points of inherent rigidity of the reservoir assembly. The test assembly shall be vibrated under the above conditions for a total period of 25 hours. During the testing, the reservoir shall be rocked back and forth at between 16 and 20 cpm, through a total angle of 30 degrees (approximately 15 degrees each side of the level position) around an axis parallel to what would be the longitudinal axis of the aircraft fuselage. If any straps or mounting members fail during the 25-hour test and require replacement, the run shall be started over again. At the conclusion of the vibration test, the reservoir shall successfully withstand the proof pressure test. When empty, the interior of the reservoir shall be examined completely for any indications of failure. If it is necessary to cut open the reservoir for complete inspection, this must be done after endurance testing (4.6.10).

4.6.10 *Endurance test.* All reservoirs which are pressurized and surged, or nonpressurized and surged shall be subjected to a pressure cycling endurance test with pressure excursion limits ranging between zero and maximum design operating pressure for a number of cycles which is comparable to twice the maximum number expected to be encountered in service, after which the reservoir shall withstand proof pressurization.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Unless otherwise specified in the contract or purchase order (see 6.2), each reservoir shall be packaged in accordance with Method III of MIL–P–116. The reservoir shall be prepared for delivery in accordance with MIL–P–7936 by securely cushioning, blocking, or bracing within a unit container conforming to PPP–B–566, PPP–B–576, PPP–B–601, PPP–B–621, PPP–B–636, PPP–B–640, or PPP–B–676.

5.2 Packing.

5.2.1 *Level A.* Components, preserved and packaged as specified in 5.1 where the unit container is not an acceptable shipping container, shall be packed in exterior shipping containers conforming to PPP–B–601 (overseas type) or PPP–B–621. Plywood shall conform to type I or II, class 2 of NN–P–515. Shipping containers conforming to PPP–B–621 shall have case liners conforming to MIL–L–10547 and appendix thereto. Box closures shall be as specified in the appendix to the applicable box specification.

5.2.2 *Level B.* Components, preserved and packaged as specified in 5.1 where the unit container is not an acceptable shipping container, shall be packed in exterior shipping containers conforming to PPP–B–591, PPP–B–621, PPP–B–585 (class 1), PPP–B–636, PPP–B–601 (domestic type), or PPP–B–640. Closures shall be as specified in the applicable box specification or appendix thereto. Fiberboard boxes shall conform to the special requirements of the applicable box specification and shall be limited to not more than 90 pounds gross weight.

5.2.3 *Level C.* Components packaged as specified in 5.1 shall be packed in a manner to ensure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL–STD–129.

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The identification shall be composed of the following information listed in the order shown:

Stock No. (or other identification number as specified in the purchase document)
RESERVOIRS: AIRCRAFT, HYDRAULIC, NON-SEPARATED TYPE MIL–R–5520
Manufacturer's part No.
Name of manufacturer
Name of contractor (if different from the manufacturer)
Contract or order No.

6. NOTES

6.1 Intended use. The hydraulic reservoirs are intended for use in aircraft hydraulic systems conforming to MIL–H–5440 to store and provide hydraulic fluid to a pump.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Type of reservoir (see 1.2).
(c) Manufacturer's part No.
(d) Level of packing required (see 5.2).
(e) Where the preproduction test sample should be sent, the activity responsible for testing, and instructions concerning the submittal of test reports (see 3.3 and 4.3.2).
(f) Method of preservation, if other than specified (see 5.1).

6.3 Separated reservoirs. MIL–R–8931 presents the requirements for hydraulic reservoirs (separated).

Custodians:
Army—MO.
Navy—Weps.
Air Force—(11).

Preparing activity:
Air Force—(11).
Project No. 1650–0053.

Reviewer activity:
Army—MO.
Navy—Weps.
Air Force—(11).

User activity:
Army—
Navy—
Air Force—

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DoD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines of reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-R-5520C Reservoirs: Aircraft, Hydraulic, Non-Separated Type

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *IF "YES" IN WHAT WAY?*

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

DD FORM 1426

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